PYTHAGORAS
ENGINEERING CONSULTANCY & SERVICES
INNOVATION FOR BETTER

5-DAY ADVANCED WORKSHOP FOR INDUSTRY PROFESSIONALS · ANSYS + PY-FFS

DESIGN BY ANALYSIS & FITNESS-FOR-SERVICE

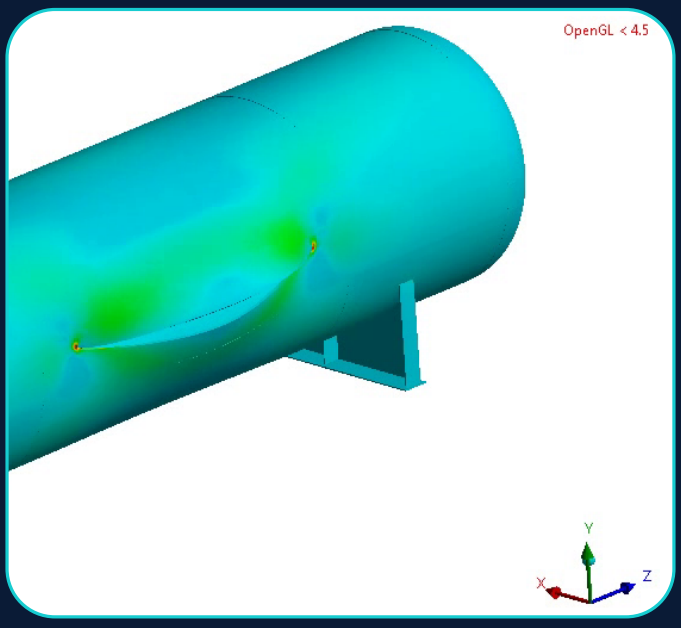
FEA with ANSYS, ASME Section VIII Division 2 Part 5 and API 579-1/ASME FFS-1 Part 9 (crack-like flaws) Levels 1, 2 and 3, for pressure vessels, storage tanks, piping and oil & gas equipment

9–13 December 2026
Wed–Sun · Bengaluru + Live online

35 contact hours

10 hours of labs in ANSYS and PY-FFS

6 months of online course access, free



3D crack growth in a pressure vessel shell

Organised by Pythagoras Engineering Consultancy and Services Private Limited
www.fea-training.com

WHY FITNESS-FOR-SERVICE

An inspection finds a crack. Run, repair or replace?

01
Is the flaw acceptable today?
Acceptance · FAD

02
How fast will it grow in service?
Crack growth and remaining life

03
When should we inspect next?
Inspection planning

Design by analysis (ASME VIII-2 Part 5) proves equipment safe; fitness-for-service (API 579-1) keeps it safely in service. **This workshop teaches both.**

WHO SHOULD ATTEND

Built for engineers who make integrity decisions

Static equipment engineers
Pressure vessels, columns, exchangers, tanks and piping

Inspection and NDE engineers
Turning flaw sizes from UT, PAUT and TOFD into decisions

Asset integrity and reliability
Life extension, RBI and turnaround planning teams

Design and FEA engineers
Stress analysts adding fracture mechanics to their toolkit

Consultants and third-party inspectors
Reviewing or preparing FFS reports for clients

Built around plant practice
Examples from vessels, tanks, piping and oil and gas equipment, with flaw sizes from UT, PAUT and TOFD turned into decisions

THE WORKSHOP AT A GLANCE

From design by analysis to fitness-for-service in five days

5 days, Wednesday to Sunday

35 contact hours

10 hours of labs in ANSYS and PY-FFS

6 months of online course access, free

Hybrid: join in the classroom in Bengaluru or live online · Certificate of participation · Small batch for individual attention

LEARNING OUTCOMES

By Sunday evening you will be able to

- ✓ Build and check FEA models of pressure equipment in ANSYS
- ✓ Linearise and categorise stresses to ASME VIII-2 Part 5
- ✓ Check plastic collapse, local failure, buckling and ratcheting
- ✓ Evaluate fatigue, including welded joints
- ✓ Assess crack-like flaws to API 579-1 Part 9, Levels 1, 2 and 3
- ✓ Estimate remaining life and inspection intervals in PY-FFS

SYLLABUS ROADMAP

Five days, one complete path

DAY 1
FEA for pressure equipment with ANSYS
Wed 9 Dec · ANSYS

DAY 2
Part 5: plastic collapse and local failure
Thu 10 Dec · ASME VIII-2

DAY 3
Part 5: buckling, fatigue and ratcheting
Fri 11 Dec · ASME VIII-2

DAY 4
API 579-1 Part 9: Levels 1 and 2
Sat 12 Dec · API 579-1

DAY 5
Level 3, crack growth and remaining life
Sun 13 Dec · API 579-1

Days 1–3 build the design-by-analysis toolkit in ANSYS; Days 4–5 apply it to crack-like flaws in PY-FFS. Each day runs 08:30–17:30 (Day 1) or 09:00–17:30 (Days 2–5), IST, and ends with a recap quiz and open questions. Day 5 closes with an assessment test, certificates and feedback.

TECHNICAL SCOPE

What is covered, and on which day

Capability	Framework	D1	D2	D3	D4	D5
FEA modelling: meshing, convergence, nonlinear, thermal, contact	ANSYS	●				
Stress categories (Pm, PL, Pb, Q, F) and linearisation	ASME VIII-2 Part 5		●			
Limit-load and elastic-plastic collapse with load factors	ASME VIII-2 Part 5		●			
Protection against local failure (strain limit)	ASME VIII-2 Part 5		●			
Buckling: eigenvalue and nonlinear with imperfections	ASME VIII-2 Part 5			●		
Fatigue: elastic, elastic-plastic, structural stress method for welds	ASME VIII-2 Part 5			●		
Ratcheting and shakedown, thermal stress ranges	ASME VIII-2 Part 5			●		
Flaw characterisation from NDE data	API 579-1 Part 9				●	
Level 1 screening and Level 2 FAD for every crack case	API 579-1 Part 9				●	
Residual stress, Kmat from Charpy data, Master Curve	API 579-1 Part 9				●	
Level 3: material-specific FAD (B), J-integral from FEA (C), ductile tearing (D)	API 579-1 Part 9					●
Crack growth (Paris law, threshold, environment), remaining life, inspection intervals	API 579-1 Part 9					●

01

DAY 1 · WEDNESDAY 9 DECEMBER · FEA

FEA for pressure equipment with ANSYS

Time	Session	Format
08:30–09:15	Registration, ANSYS Student and PY-FFS installation check	**Arrival**
09:15–10:45	FEA fundamentals for pressure equipment: elements, meshing, boundary conditions, loads and checking results	**Lecture**
11:00–12:30	ANSYS Workbench step by step: geometry, materials, mesh, supports, pressure, solution and mesh convergence	**Demo**
13:30–15:00	Linear, nonlinear (material and geometric) and thermal analysis; symmetry, contact and modelling of shells, heads and nozzles	**Lecture + demo**
15:15–17:00	Lab 1 · ANSYS: cylindrical vessel with a nozzle under pressure; mesh study and comparison with hand calculations	**Hands-on lab**
17:00–17:30	Recap quiz and open questions	**Discussion**

All times IST

02

DAY 2 · THURSDAY 10 DECEMBER · ASME VIII-2

Part 5: plastic collapse and local failure

Time	Session	Format
09:00–09:15	Recap of Day 1	**Discussion**
09:15–10:45	Design by analysis versus design by rule; the failure modes of Part 5; loads and load case combinations	**Lecture**
11:00–12:30	Elastic stress analysis: stress categories (Pm, PL, Pb, Q, F), stress linearisation and allowable limits	**Lecture**
13:30–15:00	Limit-load and elastic-plastic analysis with load factors; protection against local failure (strain limit)	**Lecture + demo**
15:15–17:00	Lab 2 · ANSYS: linearise stresses at a nozzle-to-shell junction, check the categories; elastic-plastic collapse check	**Hands-on lab**
17:00–17:30	Recap quiz and open questions	**Discussion**

ASME BPVC Section VIII, Division 2, Part 5 (Design by Analysis) · All times IST

03

DAY 3 · FRIDAY 11 DECEMBER · ASME VIII-2

Part 5: buckling, fatigue and ratcheting

Time	Session	Format
09:00–09:15	Recap of Day 2	Discussion
09:15–10:45	Buckling: the three analysis types, design factors, eigenvalue versus nonlinear analysis with imperfections	Lecture + demo
11:00–12:30	Fatigue: screening, elastic and elastic-plastic methods, the structural stress method for welds, cumulative damage	Lecture
13:30–15:00	Ratcheting and shakedown: elastic and elastic-plastic ratcheting assessment, thermal stress ranges	Lecture + demo
15:15–17:00	Lab 3 · ANSYS: buckling of a vessel under external pressure; fatigue assessment of a nozzle weld	Hands-on lab
17:00–17:30	Recap quiz and open questions	Discussion

ASME BPVC Section VIII, Division 2, Part 5 (Design by Analysis) · All times IST

04

DAY 4 · SATURDAY 12 DECEMBER · API 579-1

API 579-1 Part 9: Levels 1 and 2

Time	Session	Format
09:00–09:15	Recap of Day 3	Discussion
09:15–10:45	Part 9 procedure, data needed and Levels 1–3; flaw characterisation from NDE	Lecture
11:00–12:30	The FAD for every crack case: plates, plates with holes, cylinders, spheres, nozzles, ring-stiffened cylinders, bars and bolts, fillet welds; surface, embedded and through-wall flaws	Lecture
13:30–15:00	Stress and toughness inputs: primary, secondary and weld residual stress; Kmat from Charpy data and the Master Curve	Lecture + case
15:15–17:00	Lab 4 · PY-FFS: Level 1 screening and Level 2 FAD assessments for plate, cylinder, sphere and nozzle flaws, using worked examples	Hands-on lab
17:00–17:30	Recap quiz and open questions	Discussion

API 579-1/ASME FFS-1 Part 9 · All times IST

05

DAY 5 · SUNDAY 13 DECEMBER · API 579-1

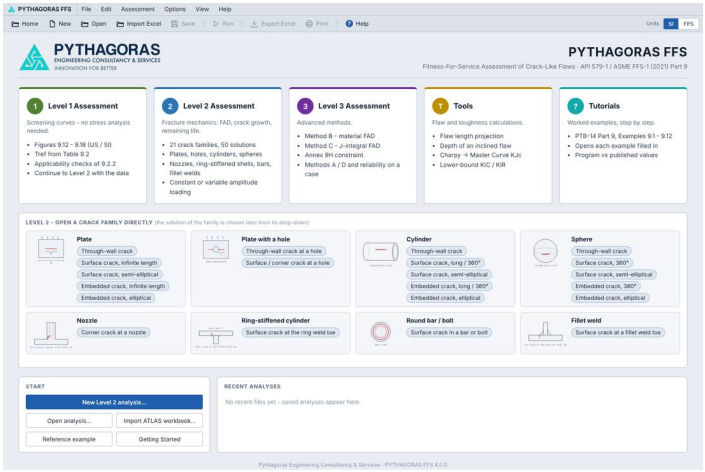
Level 3, crack growth and remaining life

Time	Session	Format
09:00–10:45	Part 9 Level 3: material-specific FAD (Method B), J-integral from FEA (Method C), ductile tearing (Method D), partial safety factors and sensitivity	**Lecture + PY-FFS demo**
11:00–12:30	Crack growth and remaining life: Paris law, threshold, environment; inspection intervals; oil and gas case studies	**Lecture + case**
13:30–15:00	Lab 5 · PY-FFS: crack growth and remaining life for plate, cylinder and sphere flaws; Level 3 material-specific FAD	**Hands-on lab**
15:15–16:30	Group project in PY-FFS: a complete assessment from inspection data to report, and team presentations	**Hands-on project**
16:30–17:00	Assessment test	**Test**
17:00–17:30	Certificates, online course access, feedback and closing	**Closing**

API 579-1/ASME FFS-1 Part 9 · All times IST

HANDS-ON LABS EVERY AFTERNOON

ANSYS on Days 1–3, our PY-FFS on Days 4–5



PY-FFS home screen with Level 1, Level 2, Level 3, Tools and Tutorials

Lab 1	**ANSYS** · vessel and nozzle model, mesh study
Lab 2	**ANSYS** · stress linearisation and plastic collapse
Lab 3	**ANSYS** · buckling and fatigue of a nozzle weld
Lab 4	**PY-FFS** · Level 1 and 2 for plates, cylinders, spheres and nozzles
Lab 5	**PY-FFS** · crack growth, remaining life, Level 3
Project	**PY-FFS** · a complete assessment to report

Bring a Windows laptop · ANSYS Student (free) for Days 1–3 · PY-FFS provided for Days 4–5

ANSYS labs follow ASME VIII-2 Part 5. PY-FFS labs cover all Part 9 crack cases, using worked examples whose results can be compared with published values.

WORKED EXAMPLE · PTB-14 EXAMPLE 9.8 IN PY-FFS

A seam flaw in a pressure vessel under pressure cycles

PROBLEM DATA

Vessel: Ri = 60 in, t = 1 in, SA-516-70

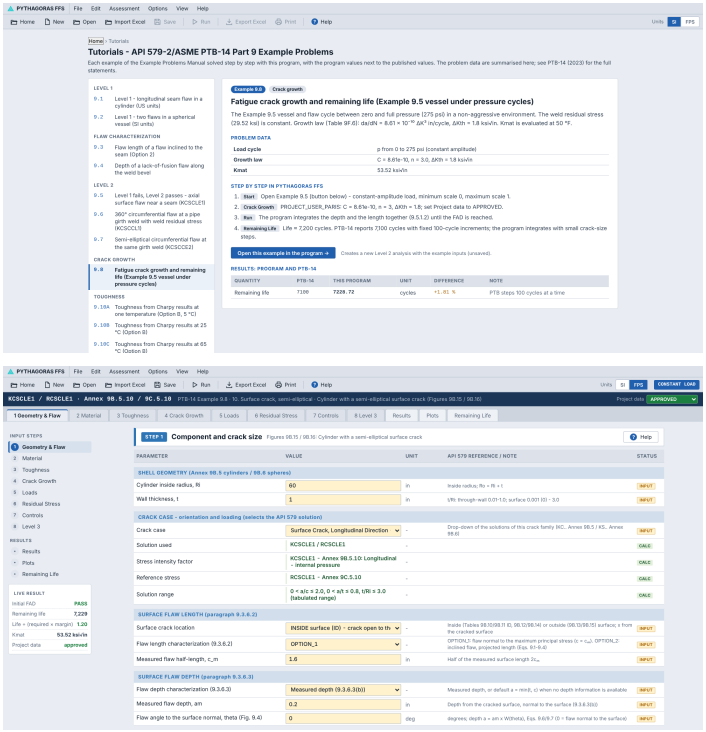
Flaw: axial surface crack at the seam, a = 0.2 in, 2c = 3.2 in

Loading: pressure cycles 0 ↔ 275 psi, weld residual stress

Growth law: da/dN = 8.61×10^{-10} ΔK^3, ΔK_{th} = 1.8 ksi√in

Toughness: Kmat = 53.52 ksi√in

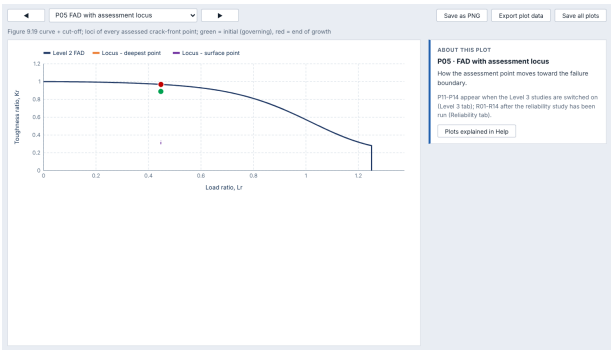
Open the example, check the inputs step by step, then run.

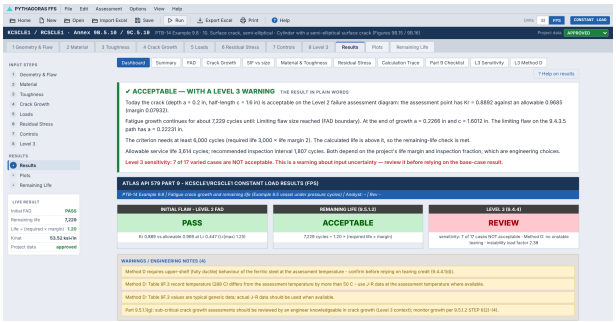


PY-FFS tutorial page with the problem statement and steps; geometry and flaw input page for the cylinder with a semi-elliptical surface crack

WORKED EXAMPLE · LEVEL 2 FAILURE ASSESSMENT DIAGRAM

Is the flaw acceptable today?





PY-FFS plot P05: Level 2 FAD with the assessment locus of the deepest and surface points · results dashboard: initial flaw PASS on the Level 2 FAD, remaining life acceptable

LR · KR

0.447 · 0.889

assessment point

ALLOWABLE KR

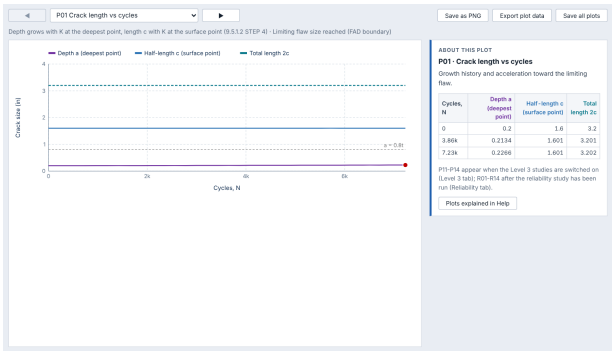
0.969 · Pass

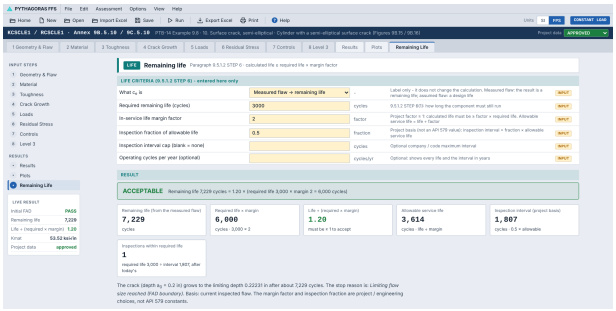
point lies inside the FAD line

The point inside the FAD line means acceptable; the locus shows it moving towards the line as the crack grows. Screenshots from PY-FFS · PTB-14 Example 9.8

WORKED EXAMPLE · CRACK GROWTH AND REMAINING LIFE

How long can it run, and when do we inspect?





PY-FFS plot P01: crack depth, half-length and total length against load cycles · remaining life page: remaining life, allowable service life and inspection interval

CYCLES TO LIMITING FLAW

7,229

cycles to the limiting flaw (PTB-14: about 7,100)

ALLOWABLE SERVICE LIFE

3,614

cycles allowable service life (margin 2)

INSPECTION INTERVAL

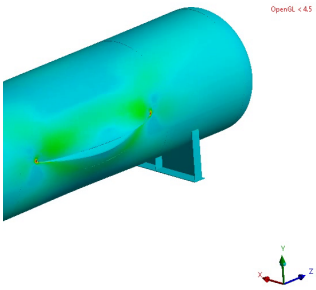
1,807

cycles inspection interval (half the allowable life)

Screenshots from PY-FFS · PTB-14 Example 9.8 · required life of 3,000 cycles set for the exercise

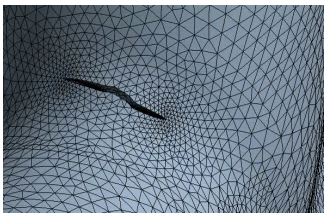
BEYOND THE HANDBOOK · 3D FRACTURE MECHANICS CASE STUDIES

When no standard solution fits: 3D fracture mechanics



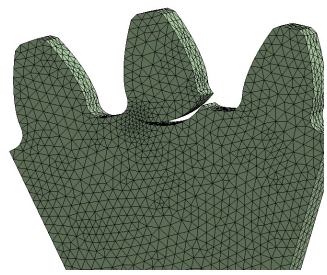
Multilayer pressure vessel

Axial crack grown through the shell; stress at both crack tips



Offshore tubular joint

Flaw at the weld hot spot, grown under wave loading



Spur gear tooth

Tooth fracture or rim fracture, decided by the rim thickness

INDUSTRIES AND EQUIPMENT COVERED

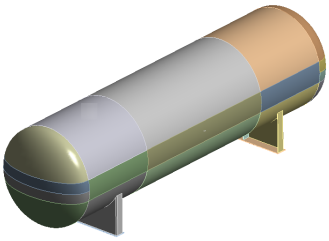
Examples from the equipment you look after

Pressure vessels and columns
Seam, nozzle and support weld flaws, life extension

Storage tanks and spheres
Flaws at shell and nozzle welds, remaining life

Piping and pipelines
Girth welds under pressure, bending and residual stress

Oil, gas and offshore
Separators, tubular joints, sour service cracking



3D model of a pressure vessel used in the Day 5 demo: horizontal vessel with shell layers, heads and saddle supports

HYBRID FORMAT

Same sessions, two ways to join

In the classroom
Inspire Workspace, Bengaluru

- Face-to-face teaching and discussion
- Lab help at your desk
- Lunch and refreshments included
- Networking with peers from industry

Live online
From anywhere, in real time (IST)

- Every session streamed live
- Questions answered during the session
- Remote support in the ANSYS and PY-FFS labs
- Session recordings for 30 days

Classroom seats are limited to keep the labs hands-on: 20 seats in Bengaluru.

DATES AND FEES

9–13 December 2026 · Wednesday to Sunday

CLASSROOM · BENGALURU

US$1,550

per participant + applicable taxes

Early bird US$1,400 until 24 November

3 or more from one company: 10% off

LIVE ONLINE

US$1,150

per participant + applicable taxes

Early bird US$1,050 until 24 November

Live sessions with remote lab support in ANSYS and PY-FFS

All fees in US dollars · Includes 6 months' online course access · Registration closes 4 December 2026

WHAT IS INCLUDED

Everything you need to practise after the workshop

Course notes
Slides, formula sheets and a step-by-step assessment checklist

Software for the labs
ANSYS Student for Days 1–3; PY-FFS for Days 4–5

Worked examples
ANSYS models and PTB-14 examples to repeat at your desk

Certificate
Certificate of participation after the assessment test

Recordings
Session recordings for 30 days (online participants)

Online course · 6 months
Free access to our online course as an additional benefit

Classroom fee also includes lunch and refreshments on all five days. Online course: www.fea-training.com

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REGISTER NOW

RESERVE YOUR SEAT

9–13 December 2026

Classroom in Bengaluru or live online. Early-bird fees until 24 November.

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Registration closes 4 December 2026